Quantum Mechanics I — Exercise Sheet 7

Angular momentum

Variant: Questions only

Conventions, as in Lecture 7: an angular momentum is a triple of self-adjoint J_a with $[J_x, J_y] = \mathrm{i}\hbar J_z$ and its cyclic permutations; $J_\pm = J_x \pm \mathrm{i}J_y$; the standard basis $|j,m\rangle$, $m = j, \dots, -j$, with $J_\pm\,|j,m\rangle = \hbar\sqrt{(j \mp m)(j \pm m + 1)}\,|j, m \pm 1\rangle$.

Exercise 1 (spin $\frac{3}{2}$).

1. Write J_z, J_+ and J_x for $j = \frac{3}{2}$ in the standard basis.
2. Check $J_+J_- - J_-J_+ = 2\hbar J_z$ on the vector $\left|\frac{3}{2}, \frac{1}{2}\right\rangle$, and find J^2.

Exercise 2 (rotations turn the triple). Let $\mathbf{J}$ be an angular momentum on $\mathbb{C}^n$ and $U_\theta = e^{-\mathrm{i}\theta J_z/\hbar}$. Show that

$$U_\theta^\dagger J_x U_\theta = \cos\theta\, J_x - \sin\theta\, J_y, \qquad U_\theta^\dagger J_y U_\theta = \sin\theta\, J_x + \cos\theta\, J_y,$$

and conclude that the vector of means $\big(\langle U_\theta\psi, J_a U_\theta\psi\rangle\big)_{a=x,y,z}$ is the vector $\big(\langle\psi, J_a\psi\rangle\big)_{a=x,y,z}$ turned by the angle θ about z.

Exercise 3 (three triples on $\mathbb{C}^4$). Three angular momenta on $\mathbb{C}^4$ have the J_z-eigenvalues

$$\text{(a) } \hbar(1,0,0,-1), \qquad \text{(b) } \hbar(\tfrac{1}{2},\tfrac{1}{2},-\tfrac{1}{2},-\tfrac{1}{2}), \qquad \text{(c) } \hbar(\tfrac{3}{2},\tfrac{1}{2},-\tfrac{1}{2},-\tfrac{3}{2}).$$

Find the multiplets in each, and say in which cases J^2 and J_z form a complete set.

Exercise 4 (the mirror is a rotation by π). For spin 1, with the matrices of Example 7.4:

1. Show that $(J_x/\hbar)^3 = J_x/\hbar$, and deduce $e^{-\mathrm{i}\theta J_x/\hbar} = \mathbb{1} - \mathrm{i}\sin\theta\,\frac{J_x}{\hbar} + (\cos\theta - 1)\frac{J_x^2}{\hbar^2}$.
2. Show that $e^{-\mathrm{i}\pi J_x/\hbar} = -M$, where M is the mirror of Lecture 6: $Me_\pm = e_\mp$, $Me_0 = e_0$.

Exercise 5 (a turned spin 1 in a magnet). Prepare the spin-1 state $|1,1\rangle$ and turn it by $U = e^{-\mathrm{i}\theta J_y/\hbar}$. A Stern–Gerlach magnet along z then reads J_z. Show that the readings $\hbar$, 0, $-\hbar$ have the probabilities

$$\left(\frac{1+\cos\theta}{2}\right)^2, \qquad \frac{\sin^2\theta}{2}, \qquad \left(\frac{1-\cos\theta}{2}\right)^2.$$

Find them at $\theta = \pi/2$, and the mean of J_z.

Exercise 6 (Kramers pairs of a spin $\frac{3}{2}$). Take one multiplet of spin $\frac{3}{2}$ with its standard basis, the time reversal Θ of Definition 7.6, and, for real d and e,

$$H = \frac{d}{\hbar^2}J_z^2 + \frac{e}{\hbar^2}\big(J_x^2 - J_y^2\big).$$

1. Write $\Theta\left|\frac{3}{2}, m\right\rangle$ for the four m, and check that $\Theta^2 = -\mathbb{1}$.
2. Show that $\Theta H = H\Theta$, that H maps the planes spanned by $\left|\frac{3}{2},\frac{3}{2}\right\rangle, \left|\frac{3}{2},-\frac{1}{2}\right\rangle$ and by $\left|\frac{3}{2},\frac{1}{2}\right\rangle, \left|\frac{3}{2},-\frac{3}{2}\right\rangle$ into themselves, and find its levels. Which vector of the second plane has the same energy as a given eigenvector of the first?
3. Add a field along z: $H_b = H + (b/\hbar)J_z$ with b real. Find its four levels, and check that for $d = 1$, $e = \frac{1}{3}$, $b = \frac{1}{10}$ they are distinct. Then, for $d \neq 0$ and $e = 0$, find a nonzero field at which two of them coincide anyway.